

Cross-Profile Seed-Tree Reuse in ReSolveD- α : Recovery of an Equivalent Signing Witness and Fresh-Message Forgery from Same-Key, Same-Message Deterministic S/F Signatures at One Security Level

Martin Feussner
Selmer Center, University of Bergen
`martin.feussner@uib.no`

1 October 2026

Provenance and review status. The ReSolveD- α attack, derivation, implementation, experiments, and initial manuscript were produced by OpenAI Codex using Daybreak Blue at Ultra reasoning effort during an AI-run audit of NGCC signature candidates. The mechanism was found after the same audit had identified a related cross-variant failure in Galas; no human first proposed its application to ReSolveD- α . A separately instructed hostile-review agent rebuilt the recovery, tried to disprove the result, and found that the conditional claim survived that review. At the date above, no human cryptanalyst has independently verified the argument, implementation, measurements, or novelty assessment. This report is a preliminary disclosure for human review and reproduction.

Abstract

ReSolveD- α supplies short S and fast F profiles at each of four security parameters. At a fixed level they use the same RSD public relation, the same untagged key formats, and a deterministic signing mode expressly defined by $\rho = 0^\lambda$. We show a practical cross-profile composition attack enabled by absent cryptographic separation of S/F keys and derivation domains under same-key reuse. At one fixed level, it applies when the same full secret key—and hence the same effective RSD witness material—signs the same message once under S and once under F with equal ρ . The deterministic mode supplies that equality. The specification inserts no profile identifier in either

$$(\mu, s) = \text{MsgHash}(\text{pk}, M), \quad (r, iv) = \text{CoinHash}(\text{sk}, \mu, \rho),$$

so the two profiles begin from the same root seed, tree salt, and IV.

The F seed tree is much smaller. Every parent of an F leaf has the same seed as the S node at the corresponding heap index, although the F and S leaf seeds themselves generally differ because the profiles use different expansion rules there. If the S opening permits reconstruction of the parent of one F-hidden leaf, the public F bottom rule recovers that leaf. Completing any one F VOLE instance gives u_a ; for $a = 0$ this is u_0 , while for $a > 0$ the serialized correction gives $u_0 = u_a \oplus c_a$. The signature then exposes the compact RSD witness

$$W = d \oplus u_0[0..\ell_{\text{wit}}).$$

This W is an equivalent signing witness and a complete signing capability: decoding it yields a regular error e with $He = y$, so an attacker can generate an honest proof on a fresh message without recovering the original secret seed.

End-to-end experiments recovered the exact honest witness and produced fresh-message S signatures accepted by verifiers compiled from untouched submitted verification sources at 160, 256, 384, and 512 bits. The lead harness’s complete fixed runs took 8.69–102.10 seconds and at most 34 MiB on an AMD EPYC host; the variation is dominated by transcript grinding. A separate public-input-only 160-bit implementation recovered in 3.99 seconds and forged in 1.92 seconds. A 256-key/message real-signer sweep recovered and forged in 256/256 cases, and all 256 fresh forgeries were accepted by the untouched verifier. Wrong-message, wrong-key, different-message, different- ρ , F-only, and mutated-witness controls failed as expected.

The result requires the same security level, the same full secret key and effective RSD witness material, the same message, and equal ρ across S and F. The deterministic mode makes equality of ρ automatic. The specification does not instruct users to reuse S/F keys, but it neither cryptographically prevents that reuse nor separates the S/F key namespaces. This report does not claim an isolated single-profile, single-oracle EUF-CMA break or an attack on independently typed S/F keys.

1 Introduction

ReSolveD- α is a code-based Fiat–Shamir signature that proves knowledge of a regular syndrome-decoding (RSD) witness using VOLE-in-the-Head, QuickSilver, and a batch all-but-one vector commitment (BAVC) [1, 2]. Each recommended security level has an S profile with shorter signatures and an F profile intended for faster signing. The profiles retain one RSD relation while changing the number and sizes of the VOLE seed trees.

That performance tradeoff creates two different views of the same deterministic seed tree. At one fixed level, equal full-key material, message, and ρ give equal roots because the key and coin derivations omit the S/F label. The small F tree switches to its bottom two-hash expansion while the much larger S tree is still in its upper correlation-robust expansion. An F leaf is therefore not itself an S leaf with the same value. Its parent, however, is a shared internal node. An S opening that reconstructs that parent completes the one leaf hidden by the F opening.

The consequence is stronger than transcript malleability. One completed F VOLE instance reveals the mask applied to the RSD witness. The recovered witness is publicly checkable and is sufficient to create fresh-message signatures. No RSD decoding, hash inversion, Fiat–Shamir fixed point, or weakness in the symmetric primitives is required.

Our contributions are as follows.

1. We prove the exact common-node identity and check the required heap bounds for every recommended S/F pair.
2. We give a public recovery algorithm that converts one complementary S/F pair into an equivalent RSD signing witness and validates it using only the public key.
3. We reproduce exact witness recovery and fresh-message forgery at all four levels, with final acceptance by untouched submitted verifiers and adversarial negative controls.
4. We separate an exact unconditioned index calculation from both the opening-conditioned geometry and a 256-key/message real-signer sweep.
5. We delimit the result as a same-level cross-profile attack requiring the same full key/effective witness material, message, and ρ . It is not a claim against an isolated S or F signing oracle.

The finding is a practical cross-profile composition attack enabled by absent cryptographic separation of S/F keys and derivation domains under same-key reuse. The specification neither instructs cross-profile key reuse nor cryptographically prevents it or separates the S/F namespaces. The submitted signer omits part of the specified `CoinHash` input; we repaired

that deviation before the main experiments, and the attack survives the repair. A separate artifact-risk control deliberately reset the untouched S/F APIs' DRNG state to induce equal nonzero ρ . That test shows that the attack is not introduced by the repaired wrapper; it is not a claim about normal randomized API use.

2 Relevant construction

2.1 Keys and the equivalent witness relation

For a fixed level, let $H = [I_{m-k} \mid H_B] \in \mathbb{F}_2^{(m-k) \times m}$. The RSD block size is six and $w = m/6$. Key generation samples two independent λ -bit strings and computes

$$\begin{aligned} \text{seed}_{pk}, \text{seed}_{sk} &\leftarrow \$ \{0, 1\}^\lambda, & H_B &= \text{SampleMatrix}(\text{seed}_{pk}), \\ e &= \text{SampleNoise}(\text{seed}_{sk}), & y &= He, \\ \text{pk} &= (\text{seed}_{pk}, y), & \text{sk} &= (\text{seed}_{pk}, \text{seed}_{sk}). \end{aligned} \tag{1}$$

Every consecutive six-bit block of e has Hamming weight one.

Write $e = (e_A, e_B)$ according to the systematic split. The compact witness is $W = \text{EncodeNoise}(e)$. It stores the first five bits of each six-bit block of e_B , and the omitted bit restores odd parity. Given (pk, W) , anyone can reconstruct e_B and compute

$$e_A = y \oplus H_B e_B. \tag{2}$$

Consequently, any W for which every reconstructed block of e has weight one is an equivalent signing witness. It need not lie in the image of **SampleNoise**, and recovering seed_{sk} is unnecessary.

The four recommended RSD tuples and profile sizes are shown in Table 1. At one level, S and F share (m, k, w) , the serialized key format, and the public witness relation. Independent **KeyGen** calls still sample independent keys; the attack assumes reuse of the same raw full secret-key material across the two profiles.

Table 1: Recommended ReSolveD- α parameters used by the attack.

λ	(m, k, w)	τ_S	τ_F	L_S	L_F	$ W $ bytes	$ \sigma_S $	$ \sigma_F $
160	(1860, 1056, 310)	14	21	28,672	3,328	110	5,307	6,851
256	(2988, 1698, 498)	22	35	49,152	4,864	177	13,973	17,932
384	(4452, 2532, 742)	34	53	69,632	7,296	264	31,575	40,469
512	(5940, 3378, 990)	46	72	94,208	9,216	352	56,239	72,611

2.2 Signing state and witness masking

For message M , the specified signer derives

$$(\mu, s) = \text{XOF}(0x00 \parallel \text{seed}_{pk} \parallel y \parallel M; 3\lambda), \tag{3}$$

$$(r, iv) = \text{XOF}(0x01 \parallel \text{sk} \parallel \mu \parallel \rho; \lambda + 120). \tag{4}$$

The signer normally samples $\rho \leftarrow \$ \{0, 1\}^\lambda$; Section 4.2.2 of the submission expressly defines deterministic signing by setting $\rho = 0^\lambda$. Neither equation contains the profile identifier.

The profile-specific BAVC and VOLE algorithms expand (r, s, iv) into instances $u_0, \dots, u_{\tau-1}$. The signature serializes corrections

$$c_a = u_0 \oplus u_a \quad (a \geq 1) \tag{5}$$

and masks the RSD witness as

$$d = W \oplus u_0[0..\ell_{\text{wit}}]. \quad (6)$$

The BAVC opening reveals every seed except one leaf per VOLE instance, while the hidden leaf commitments remain serialized. Fiat–Shamir hashes bind the commitment, corrections, d , and QuickSilver proof into the final challenge.

Verification reconstructs the opened VOLE view, recomputes all three Fiat–Shamir challenges, and checks the QuickSilver proof for the public RSD relation. The proof coins are not recomputed from seed_{sk} . Therefore an attacker who knows any valid W can execute the public prover with attacker-chosen coins and make a conventional fresh-message signature.

2.3 The pcGGM heap

A profile with L physical leaves uses a one-based heap with root R_1 , internal nodes $R_1, \dots, R_{L-1}$, and leaves $R_L, \dots, R_{2L-1}$. For public (s, iv) its upper expansion is

$$\begin{aligned} R_{2i} &= \text{TCCR}(R_i, s, iv), \\ R_{2i+1} &= R_{2i} \oplus R_i, \end{aligned} \quad i < L/2, \quad (7)$$

and its bottom expansion is

$$\begin{aligned} R_{2i} &= \text{TCCR}(R_i \oplus \text{const}_0, s, iv), \\ R_{2i+1} &= \text{TCCR}(R_i \oplus \text{const}_1, s, iv), \end{aligned} \quad L/2 \leq i < L. \quad (8)$$

The final challenge selects one hidden leaf in every logical VOLE instance. The opening serializes a minimum span of node seeds that covers all other leaves. It therefore permits reconstruction of precisely the nodes outside the union of paths from the hidden leaves to the root.

3 The cross-profile identity

Lemma 1 (Shared deterministic state). *Fix one level, a key pair (pk, sk) , and a message M . S and F signing with the same ρ produce identical (μ, s, r, iv) . In particular, this holds for the specified deterministic mode $\rho = 0^\lambda$.*

Proof. At a fixed level the profiles have byte-identical key formats and public RSD parameters. Equations (3) and (4) are deterministic functions of the displayed values and contain no S/F identifier. Their inputs and outputs are therefore equal. $\square$

Lemma 2 (Shared F internal nodes). *Under Lemma 1,*

$$R_i^F = R_i^S \quad (1 \leq i < L_F).$$

An F leaf seed is generally not equal to the S seed at the same heap index.

Proof. The roots agree. Suppose $1 < i < L_F$. The parent that generates R_i has index strictly below $L_F/2$. Both profiles therefore apply the upper rule in (7) at that parent, with the same seed, s , and iv . Induction proves equality of every F internal node.

For an F leaf, its parent lies at or below the point where F switches to the bottom rule (8), while S is still using the upper rule. The two child computations differ, so equality of the leaf seeds does not follow. $\square$

Proposition 3 (All F leaf parents are usable S nodes). *For every recommended pair, every F leaf position is an internal S heap position, and the seed of its parent is shared by Lemma 2.*

Proof. In zero-based notation the F leaves occupy positions $L_F - 1, \dots, 2L_F - 2$, while the last S internal position is $L_S - 2$. Table 2 checks the stronger comparison requested directly from the submitted schedules:

$$2L_F - 2 < L_S - 1.$$

Moreover, an F leaf parent has one-based index in $[L_F/2, L_F - 1]$. Its value is an F internal seed and hence shared by Lemma 2; each listed $L_F - 1 < L_S/2$, so S has not reached its bottom rule at that parent. $\square$

Table 2: Self-contained heap bounds for all recommended profile pairs.

Pair	$2L_F - 2$	$L_S - 1$	$L_F - 1$	$L_S/2$
160S/F	6,654	28,671	3,327	14,336
256S/F	9,726	49,151	4,863	24,576
384S/F	14,590	69,631	7,295	34,816
512S/F	18,430	94,207	9,215	47,104

The distinction in Lemma 2 is important. The attack does not reinterpret an S node as an equal F leaf. It asks the S opening to reconstruct the shared *parent* seed and then applies the public F bottom rule to that seed.

4 Equivalent signing witness recovery and forgery

Let h_a be the F-hidden physical leaf for logical instance a , and let $p_a = \lfloor h_a/2 \rfloor$. Decode both final challenges and reconstruct the nodes permitted by each opening. If p_a is outside the S hidden-path union, the S opening permits reconstruction of $R_{p_a}^S$. By Lemma 2, this is $R_{p_a}^F$. Applying the left or right branch of (8) gives the missing F leaf seed.

The F opening already reconstructs every other leaf seed in instance a . Public PRG expansion and XOR therefore give the complete u_a . Write N_a for the number of F leaf seeds contributing to that instance, $\hat{\ell}$ for the bit length expanded from each seed, and b for the number of output bits charged to one underlying PRG primitive block. Equation (5) gives

$$u_0 = \begin{cases} u_a, & a = 0, \\ u_a \oplus c_a, & a > 0, \end{cases} \quad (9)$$

and the witness follows from Equation (6):

$$W = d \oplus u_0[0..\ell_{\text{wit}}]. \quad (10)$$

Theorem 4 (Conditional recovery of an equivalent signing witness). *Suppose, at one fixed security level, the same ReSolveD- α full secret key—and hence the same effective RSD witness material—signs one message under S and F with equal ρ . If at least one F hidden-leaf parent lies outside the S hidden-path union, the two public signatures determine an equivalent RSD signing witness in $O(L_S + L_F + N_a \lceil \hat{\ell}/b \rceil)$ primitive-block work and $O((L_S + L_F)\lambda)$ bits of memory.*

Proof. The equal signing state follows from Lemma 1. For the available instance, Proposition 3 and the S opening reconstruct the shared parent, and the F rule reconstructs the missing leaf. All inputs to the PRG expansions, corrections, and Equations (9)–(10) are public, so they recover W without secret state. Decoding W with Equation (2) and checking each weight-one block validates the public relation. Traversing the trees costs $O(L_S + L_F)$ primitive work, and expanding the N_a F leaf seeds costs $O(N_a \lceil \hat{\ell}/b \rceil)$ primitive blocks. Streaming those leaf images while storing the two seed heaps uses $O((L_S + L_F)\lambda)$ bits. $\square$

The complete chain is

$$\boxed{\begin{array}{l} \text{missing S/F separation} \implies \text{equal roots and shared parents} \\ \implies \text{complementary opening} \implies u_a \implies u_0 \implies W \implies \text{fresh-message proof.} \end{array}}$$

The public validation step is material. The attacker reconstructs e from $(\mathbf{pk}, W)$, checks every six-bit block has weight one, and checks $He = y$. A successful pair is therefore recognizable without the original seed_{sk} . To forge, the attacker runs the ordinary prover on a fresh message using W and attacker-chosen proof coins. The verifier checks the public relation and has no way to require that W came from the original noise-sampling seed.

5 Success probability and retry strategy

For a fixed F parent p , let

$$c_b(p) = \#\{j : p \text{ is an ancestor of the S physical leaf for } (b, j)\}.$$

If the decoded S indices are independent and uniform before grinding and opening-cap rejection, the exact probability that the S opening reconstructs p is

$$P_p = \prod_{b=0}^{\tau_S-1} \left(1 - \frac{c_b(p)}{N_{S,b}}\right). \quad (11)$$

Averaging Equation (11) over the uniform hidden index of F instance zero gives the second column of Table 3.

Table 3: Analytic and empirical opening geometry. The analytic column is an unconditioned independent-index model, not an exact serialized-signature probability.

Pair	analytic $\Pr[p_0 \text{ exposed}]$	conditioned p_0	conditioned any instance
160S/F	0.991823130948	992/1,000	1,000/1,000
256S/F	0.991696001787	988/1,000	1,000/1,000
384S/F	0.991401471498	987/1,000	1,000/1,000
512S/F	0.990043848760	990/1,000	1,000/1,000

The qualification is essential. A real signer serializes only challenges whose proof-of-work suffix is zero and whose minimum BAVC opening fits T_{open} . Those conditions alter the index distribution, and S/F challenges are generated from related transcripts. The last two columns come from 1,000 independently sampled challenge pairs per level after applying the exact submitted opening-size condition. They are empirical conditioned counts, not proofs of the real-signer probability.

The real event is publicly testable: attempt recovery for each F instance and validate the resulting W against the public RSD relation. If no parent is available, request another deterministic S/F pair on a new common message. Changing the message changes μ and the root except with negligible hash collision probability.

We also tested the complete signer rather than only its decoded-index geometry. Across 256 distinct deterministic test keys and messages at 160 bits, recovery and fresh-message forgery succeeded in 256/256 cases. Instance zero was enough in 255 cases; the remaining case succeeded with instance one. This experiment supports practical retry behavior, but it does not turn the unconditioned model into an exact probability theorem.

6 End-to-end experiments

6.1 Specification repairs and verifier separation

The official specification, not accidental source behavior, defines the design. The main experiment began from the submitted reference implementation and made one signer conformance repair: the source’s `hash_r_iv` absorbed only seed_{sk} , whereas Equation (4) specifies the full $\text{sk} = \text{seed}_{pk} \parallel \text{seed}_{sk}$. The repaired signer inserts seed_{pk} immediately before seed_{sk} . It does not add an S/F identifier and therefore preserves the cross-profile equality.

Two additional helper functions expose public operations already performed by verification: reconstructing a permitted BAVC node and XORing the PRG images of a completed VOLE instance. They receive only serialized signatures, public parameters, public keys, and messages. They do not read the secret key, honest witness, root, or retained signer state.

The submission also contains non-executable or contradictory notation for the QuickSilver packing, VOLE row length, and byte padding. The experiment uses the source-consistent intended transcript and records those interpretations in the reproducer. The attack does not exploit them: the recovered W equals the honest regular witness, satisfies the public RSD relation directly, and can be used with either the submitted constraint implementation or the natural repaired specification. The extraction itself stops before QuickSilver.

Final acceptance was tested by a separate executable linked directly against untouched submitted verification sources. No conformance-repaired or instrumented source file is linked into that binary. Because `CoinHash` is a signer-only coin derivation and is never recomputed by verification, its repair cannot cause a false verifier acceptance.

6.2 All-level recovery and forgery

For each recommended level, the experiment generated one S and one F signature under the same key and message with the specified deterministic $\rho = 0^\lambda$. It verified both signatures, recovered W , compared it to the independently retained honest witness, and signed a fresh message from the recovered witness with attacker-selected proof coins. The untouched submitted S verifier accepted every fresh signature and rejected that signature under the original query message.

Table 4 reports complete process measurements, including the two honest query signatures, recovery, fresh proof generation, and controls. Signing grinds both a challenge suffix and an opening-size condition, so fixed inputs vary substantially; the 512-bit case happened to accept unusually quickly and should not be read as a stable benchmark.

Table 4: Lead-harness, specification-repaired end-to-end experiments on the AMD EPYC audit host. “Fresh” and “wrong” are results from untouched submitted verifier sources.

Pair	query bytes	witness bytes	wall (s)	RSS (MiB)	fresh	wrong
160S/F	12,158	110	12.60	4.5	accept	reject
256S/F	31,905	177	36.44	10.5	accept	reject
384S/F	72,044	264	102.10	19.4	accept	reject
512S/F	128,850	352	8.69	33.7	accept	reject

A separately compiled 160-bit pipeline divided the attack into two processes. The recovery process accepted only $(\text{pk}, M, \sigma_S, \sigma_F)$, verified both inputs, recovered instance zero, and checked the public RSD relation. It took 3.99 seconds and 2,840 KiB RSS. A second process accepted only $(\text{pk}, M_{\text{fresh}}, W)$, used fixed attacker-selected proof coins, and produced a fresh S signature in 1.92 seconds and 4.0 MiB RSS. The pristine verifier accepted it. The recovered 110-byte

witness agreed with the separately retained honest witness only in a post-recovery validation comparison.

An independent hostile implementation repeated public-only recovery and forgery at all four levels. As an artifact-risk control, it also compiled the untouched submitted 160S and 160F public APIs as separate programs. After generating one submitted key, it deliberately reset the documented DRNG state before the two signing calls, inducing the same nonzero ρ in both untouched signers. Those original-interface query signatures again yielded the equivalent witness, and the fresh forgery was accepted by a separately compiled pristine verifier. This control establishes wrapper independence only. The untouched API normally samples fresh ρ , so the control is not evidence of a vulnerability in normal randomized use. The hostile run’s largest measured resident set across the four levels was 46.8 MiB; this is separate from the lead-harness measurements in Table 4.

6.3 Real-signer sweep and negative controls

The 256-case ReSolveD- α -160 sweep used distinct deterministic test keys and messages. Every case checked honest S/F verification, IV equality, exact witness recovery, public-relation validity, fresh-message proof generation, and wrong-message rejection. All 256 fresh signatures were then replayed through the untouched submitted verifier: it accepted all 256 under their fresh messages and rejected all 256 under their original query messages. Per-case wall times ranged from 12.00 to 19.36 seconds, with median 14.895 seconds; peak RSS was at most 4,752 KiB. These per-process times are contention-affected rather than single-core benchmarks.

Two additional independent 160-bit cases ran the full adversarial controls. Each control had the expected result:

- the F opening alone could not reconstruct any of its hidden-leaf parents;
- pairing F with an S transcript from a different key, different message, or different ρ produced a wrong witness and a rejected proof;
- a one-bit-mutated recovered witness produced a rejected signature;
- a one-bit-mutated signature was rejected by the pristine verifier; and
- the valid fresh forgery was rejected under the query message and an unrelated message.

The public recovery code has no secret-key or ground-truth input. Secret data is used only by the oracle processes that create the two requested signatures and by the later equality check used to validate the experiment.

6.4 Reproduction

The companion repository directory contains a deterministic reproducer. From `reproducer/`, run

```
./run.sh.
```

The command verifies the bundled source manifest, builds the conformance-repaired query signer, public recovery program, witness-based prover, and pristine verifier, then performs the 160-bit recovery, fresh-message forgery, and negative controls. A successful run ends with `minimal_reproducer=pass`.

The optional command `./run_all_levels.sh` repeats witness recovery, fresh-message forgery, and untouched-verifier acceptance at all four levels. The reproducer README states the compiler requirements, expected runtime and memory, source-archive checksum, exact conformance patch, and the distinction between the repaired signer and untouched verifier. No audit-internal path is required.

7 Security impact and limits

This is a practical cross-profile composition attack enabled by absent cryptographic separation of S/F keys and derivation domains under same-key reuse. Its demonstrated consequence is recovery of an equivalent RSD signing witness followed by a fresh-message signature accepted by the public verifier. It is a concrete recovery-and-forgery chain based on an equivalent signing witness, rather than a reduced security estimate. The original sampling seed seed_{sk} is not recovered. The recovered W is nonetheless a complete signing capability because it satisfies the same public relation proved by an honest signer.

Same-key cross-profile access. The attacker needs S and F signatures at one level under the same full secret key, and therefore the same effective RSD witness material. The specification does not instruct deployments to reuse keys across profiles. **KeyGen** takes a parameter set, but the S/F profile is not cryptographically bound into or encoded by the generated key material; both profiles accept the same serialized key format and use the same RSD relation. Thus the specification does not cryptographically prevent same-key cross-profile reuse or separate the S/F key namespaces. A deployment that generates and enforces independent typed keys for S and F avoids this prerequisite.

Same message and ρ . The two signatures must cover the same message and use the same ρ . The specified deterministic mode makes equality of ρ automatic by fixing $\rho = 0^\lambda$. The default submitted API instead normally samples fresh ρ ; absent an accidental equality, randomized S/F signatures do not share the root and this attack does not apply. The reset-DRNG experiment deliberately induced equal nonzero ρ only to test wrapper independence; it does not show a normal randomized-API vulnerability. Different messages likewise change μ and the derived state.

No isolated EUF-CMA claim. The ordinary EUF-CMA game for one fixed profile provides one signing oracle. This attack needs two profile interfaces under one key. It therefore does not establish an EUF-CMA break of ReSolveD- α -160S alone, 160F alone, or any other independently keyed single profile.

Opening geometry. The heap bounds are exact, but successful exposure of an F parent depends on the two challenges. Table 3 separates an exact unconditioned model from conditioned sampling, and the 256-case signer sweep is empirical. A failed pair is publicly recognizable and can be retried with a new common message; the experiments do not prove that every valid pair succeeds.

Specification interpretation. A literal reading of the PDF’s Lemma 3.5 packing formula makes later QuickSilver loop indices run outside the packed vector at 160, 256, and 384 bits and omits 230 constraint checks at 512 bits. The submitted source instead hashes the constraint blocks separately and does not exhibit that literal packing issue. Xiong and Wang independently flag the printed Lemma 3.5/QS.RSDProve block-index shift as a specification erratum [4]. The experiments use the source’s natural completion. The cross-profile leak occurs in the preceding BAVC/VOLE layer: the recovered string independently passes the explicit RSD relation, and fresh forgeries pass the untouched submitted verifier. The attack does not treat the malformed formula as a weakened proof relation.

8 Repair

The direct repair is to bind the complete profile identifier into the signing state before any tree expansion, for example

$$\begin{aligned}(\mu, s) &= \text{XOF}(0x00 \parallel \text{instanceID} \parallel \text{pk} \parallel M; 3\lambda), \\(r, iv) &= \text{XOF}(0x01 \parallel \text{instanceID} \parallel \text{sk} \parallel \mu \parallel \rho; \lambda + 120).\end{aligned}\tag{12}$$

Here `instanceID` must canonically include both the security level and the S/F profile. The same identifier can be included in the BAVC and Fiat–Shamir frames as defense in depth.

Keys and APIs should also be typed by the full instance. An S key should be rejected by the F signing entry point even though the underlying RSD relation is the same. Conformance tests should assert that equal raw key/message inputs in S and F yield different roots and IVs. Either independent key namespaces or root domain separation blocks the demonstrated equality; deploying both makes the intended boundary explicit.

9 Prior art and novelty boundary

ReSolveD introduced the RSD-specific VOLE-in-the-Head proof and its two performance profiles [2]. After accounting for the live sign-21-1 report below, a best-effort search of the NGCC inventory, the PKC Forum, the IACR ePrint Archive, the candidate title and authors, and combinations of “ReSolveD”, “cross-profile”, “same key”, “deterministic”, “seed tree”, and “domain separation” found no public report of the specific cross-profile witness recovery in this paper. This negative search is not proof that unpublished or unindexed work does not exist.

During finalization on 1 October 2026, Xiong and Wang, with reported GLM-5.3 assistance, published a distinct one-signature multi-target attack on ReSolveD- α ’s use of one TCCR tweak for every tree node [4, 3]. It enumerates a hidden-parent candidate once and tests it against the many revealed nodes of one signature, for an expected $2^\lambda/(T + 1)$ TCCR evaluations. Median target counts give about $2^{248.2}$, $2^{375.6}$, and $2^{503.2}$ evaluations for the affected 256-, 384-, and 512-bit sets; the 160-bit sets remain above their 128-bit target. The authors describe the resulting 7–9-bit reduction as non-practical: scaled searches validate the multi-target factor and recovery chain, but the full exponential search was not run. The submission team nevertheless confirmed the missing per-node tweak as a specification bug [5]. That attack performs exponential same-profile enumeration. The attack here instead completes a hidden F leaf directly from a shared parent reconstructed through S and runs in practical linear time. A per-node tweak that remains equal across S and F does not separate their root state, while an S/F derivation tag alone does not repair within-signature TCCR target reuse; the two fixes address different defects.

Chosen-protocol attacks give the general warning that two protocols can become insecure when they expose a common key [6]. Modified-parameter attacks similarly show that security for one parameter choice need not survive reuse of the same key under another [7]. The 2024 PQC Forum discussion of parameter-set key separation made this issue concrete for modern signatures: Robin Larrieu suggested that even one discriminator bit between the “s” and “f” SLH-DSA profiles would suffice at a fixed hash family and security level [8]. That discussion proposed preventive domain separation; it did not give the ReSolveD- α attack below.

There are also close mechanistic precedents. Bui et al. explain that reuse or collision of a GGM master seed lets a revealed co-path complete an unopened leaf and expose the masked witness [9, Sec. 3.1]. Jendral and Dubrova fault FAEST root expansion so that two VOLEitH trees coincide; when their openings omit different leaves, one tree completes the other, permitting reconstruction of u and recovery of the witness masked by d [10, Sec. 8]. Their triggers are respectively a collision search and a physical fault. Jendral–Dubrova is the closest precedent

for the tree mechanism used here. Feneuil and Rivain’s formal study of correlated GGM trees likewise makes tree correlation and domain separation explicit security conditions [11].

Delgado’s contemporary SDitH v2 attack is the closest standard-model and domain-separation precedent [12]. It also recovers a permanent witness and forges from public cross-signature transcripts whose hidden-leaf endpoints mask that witness. Its mechanism and cost are different: it uses $q = 2^{12}$ ordinary same-profile signatures, exponential seed enumeration, unary filters, and trie/Gray-code acceleration, with reported full-parameter costs about $2^{129.957}$, $2^{193.909}$, and $2^{258.068}$; its experiments use reduced domains. The attack here instead needs one identical-root S/F pair and directly completes a hidden leaf from a parent reconstructed through the complementary opening, giving a practical linear-time recovery. Delgado’s proposed salt, leaf, and block stream labels do not supply ReSolveD- α ’s missing S/F derivation tag, while an S/F tag alone would not repair Delgado’s distinct expansion-domain issue.

The Galas cross-variant attack from the same NGCC audit is the direct same-key, same-message S/F precedent [13]. The present claim is therefore deliberately narrow. Its ReSolveD-specific contribution is the no-fault trigger supplied by specified deterministic signing under unseparated S/F namespaces, together with the shared-parent/different-leaf pcGGM identity, completion of an arbitrary F VOLE instance through serialized corrections, and recovery of an equivalent RSD witness rather than the original secret seed. It does not claim a new generic seed-tree attack family.

10 Conclusion

At one fixed security level, one deterministic S/F signature pair under the same full secret key and effective RSD witness material, the same message, and equal $\rho = 0^\lambda$ can reveal an equivalent signing witness with full signing capability. Exposure is challenge-dependent; failure is publicly recognizable, and the reported deterministic 160-bit sweep succeeded in 256/256 cases. Missing profile separation makes the two executions share their root and F internal-node seeds. The S opening can reconstruct the parent of an F-hidden leaf; public F bottom expansion then completes one VOLE instance, exposes u_0 , and unmask the compact RSD witness. That witness produces ordinary fresh-message signatures.

The attack was reproduced at all four recommended levels and independently checked through untouched submitted verifiers. Its scope remains conditional: it needs the same security level, full key/effective witness material, message, and ρ across S and F, with the specified deterministic mode providing equal ρ . Typed keys and full instance domain separation remove that shared state and prevent the recovery. This is not an isolated single-profile, single-oracle EUF-CMA break.

Acknowledgements

The computations were performed on the Norwegian Research and Education Cloud (NREC), using resources provided by the University of Bergen and the University of Oslo.

References

- [1] Yu Yu, Kang Yang, Xiaojie Guo, Hongrui Cui, Chun Guo, Di Yan, Yituo He, Lizheng Wang, and Jiangxia Ge, *ReSolveD- α : Algorithm Specifications*, NGCC submission, 30 June 2026, with parameter addition dated 28 August 2026.
- [2] Hongrui Cui, Hanlin Liu, Di Yan, Kang Yang, Yu Yu, and Kaiyi Zhang, *ReSolveD: Shorter Signatures from Regular Syndrome Decoding and VOLE-in-the-Head*, Public-Key Cryptography 2024; IACR Cryptology ePrint Archive, Report 2024/040, <https://eprint.iacr.org/2024/040>.

- [3] NGCC Security Reports, *ReSolveD- α (sign-21)*, <https://ngcc.dev/reports/sign-21.html>, accessed 1 October 2026.
- [4] Zhenyu Xiong and Mingsheng Wang, *ReSolveD- α —No Per-Node Tweak in the BAVC/TCCR Tree: One Signature Gives a Multi-Target Seed Search*, PKC Forum public comment, with reported GLM-5.3 assistance, 1 October 2026, <https://list.niccs.org.cn/archives/list/pkcforum@list.niccs.org.cn/message/PH6Q6UHS7YJAN3RHER7YWUNXOHOZYIFF/>.
- [5] The ReSolveD- α Team, *Response Confirming the TCCR-Tweak Specification Bug*, PKC Forum, 1 October 2026, <https://list.niccs.org.cn/archives/list/pkcforum@list.niccs.org.cn/message/GEDCFLSHBQTVSIW5K7ACXWGY7GK0XASI/>.
- [6] John Kelsey, Bruce Schneier, and David Wagner, *Protocol Interactions and the Chosen Protocol Attack*, Security Protocols, 5th International Workshop, Lecture Notes in Computer Science 1361, pp. 91–104, Springer, 1998, <https://doi.org/10.1007/BFb0028162>.
- [7] Nick Howgrave-Graham, Joseph H. Silverman, Ari Singer, and William Whyte, *Modified Parameter Attacks: Practical Attacks Against CCA2 Secure Cryptosystems, and Countermeasures*, IACR Cryptology ePrint Archive, Report 2004/344, 2004, <https://eprint.iacr.org/2004/344>.
- [8] Robin Larrieu, message in *Official comment on FIPS 203 ipd: seed as decapsulation key*, NIST PQC Forum, 3 June 2024, https://groups.google.com/a/list.nist.gov/g/pqc-forum/c/5CT4NC_6zRI/m/lpifFrpWAJ.
- [9] Dung Bui, Eliana Carozza, Geoffroy Couteau, Dahmun Goudarzi, and Antoine Joux, *Faster Signatures from MPC-in-the-Head*, ASIACRYPT 2024; IACR Cryptology ePrint Archive, Report 2024/252, <https://eprint.iacr.org/2024/252>.
- [10] Sönke Jendral and Elena Dubrova, *Side-Channel and Fault Injection Attacks on VOLEith Signature Schemes: A Case Study of Masked FAEST*, IACR Cryptology ePrint Archive, Report 2025/378, revised 2026, <https://eprint.iacr.org/2025/378>.
- [11] Thibault Feneuil and Matthieu Rivain, *On the Security of MPC-in-the-Head Signatures with Correlated GGM Trees*, IACR Cryptology ePrint Archive, Report 2026/615, 2026, <https://eprint.iacr.org/2026/615>.
- [12] José Luis Delgado, *Cross-Signature Signing-Key Recovery and Domain-Separation Repair for SDitH v2*, IACR Cryptology ePrint Archive, Report 2026/1808, 2026, <https://eprint.iacr.org/2026/1808>.
- [13] Martin Feussner, *Cross-Variant Seed-Tree Reuse in Galas: Secret-Key Recovery and Fresh-Message Forgery from Same-Key, Same-Message S/F Signatures*, 1 October 2026, https://github.com/martinfeussner/NGCC-Signature-Audit/blob/main/Galas/Galas_Attack_Description.pdf.